5

$$
\begin{aligned}
F_{\text{on Q}} &= m_{\text{Q}}a = 4a \\
&= 4 - 0.8 - T
\end{aligned}
$$

$$
\begin{aligned}
F_{\text{on P}} &= m_{\text{P}}a = 2a \\
&= T - 0.8
\end{aligned}
$$

solve simultaneous:

$$
\begin{aligned}
4a &= 4 - 0.8 - T \\
2a &= T - 0.8 \\
a &= 0.4 \text{ ms}^{-2}
\end{aligned}
$$

motion of P relative to Q:

$$
\begin{aligned}
a_{\text{relative}} &= 2a \\
s_{\text{relative}} &= 0.40 - 0.15 = 0.25 \text{ m}
\end{aligned}
$$

$$
\begin{aligned}
s &= ut + \frac{1}{2}at^2 = 0 + \frac{1}{2}at^2 \\
t &= \sqrt{\frac{2s}{a}} = \sqrt{\frac{2(0.25)}{0.8}} \\
&= 0.79 \text{ s}
\end{aligned}
$$

6